nvfp4_gemv

Ended 2025-11-29 06:59 UTC

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▼ Description

You will implement a batched matrix-vector multiplication kernel optimized for NVIDIA B200. To be explicit, you will be given a tuple of tensors:

```
(a, b, sfa, sfb, c)
```

where:

- `a` is M x K x L in K-major order in nvfp4(e2m1)
- `b` is 1 x K x L in K-major order in nvfp4(e2m1)
- `sfa` is M x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `sfb` is 1 x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `c` is M x 1 x L in fp16

Matrix sizes `M` is divisible by mma_tiler_mn[0] defined in the kernel, `K` is divisible by 64. The ranking criteria is the geometric mean of the benchmark results. For the grand price, your kernel will be evaluated against the speed of light analysis and the solution closest to the speed of light will be awarded the grand price.

```
The speed of light analysis based on the max(FFMA math throughput, DRAM memory
M    K     L time[us]
7168 16384 1 8.622
4096 7168  8 17.275
7168 2048  4 4.317
```

▼ Benchmark Shapes

- `{"k":16384,"l":1,"m":7168}`
- `{"k":7168,"l":8,"m":4096}`
- `{"k":2048,"l":4,"m":7168}`

RANKINGS REFERENCE SUBMISSION

NVIDIA

Hide

s.am._	18.550µs		sub_final.py
gau.nernst	18.553µs	+0.003µs	submission_vmix.py
shellsmile15795	18.708µs	+0.155µs	submission_cu_v22.py
_spatters	19.582µs	+0.874µs	v5.py
netvope	19.943µs	+0.362µs	submission.py
mdouglas	20.293µs	+0.350µs	submission_new.py
lambda	21.598µs	+1.305µs	submission_c_a3.py
sazc	21.779µs	+0.180µs	submission-v2.py
aikitoria	22.072µs	+0.294µs	submission2.py
tomaszki	22.343µs	+0.271µs	submission.py
yue	22.392µs	+0.049µs	submit_try0.py
JB Gage	22.418µs	+0.026µs	submission.py
snowclipsed	22.490µs	+0.072µs	kmajor.py
v0i0	22.542µs	+0.052µs	ubmission-28-12-52-23.p
txacvalh	22.710µs	+0.168µs	good_kernel.py
Naturalseeker	22.921µs	+0.211µs	nvfp4_gemv_semi_opv.py
Nareg	23.020µs	+0.098µs	submit_final.py
Venkat Raman	23.174µs	+0.154µs	ission_transposed_cach
macto	23.301µs	+0.127µs	submission.py
cdtmc	23.767µs	+0.466µs	new.py
Samir	23.939µs	+0.171µs	submission6.py
rex_cz	24.260µs	+0.321µs	fp4_batched_gemv_v10.
guojun	24.580µs	+0.321µs	submission_v2.5.py
J	25.017µs	+0.436µs	submission.py
Simon	25.112µs	+0.096µs	new.py
tipsytoad__20249	25.292µs	+0.180µs	gemv_triton.py

zhumakhan_nazir	25.380µs	+0.088µs	template_cute.py
swanbomb_	25.989µs	+0.608µs	lversion.py
jason	26.107µs	+0.119µs	submission.py
rcmalli	26.301µs	+0.194µs	submission_cdv22.py
TumoiYorozu	26.461µs	+0.160µs	sub.py
DankFarrik	27.016µs	+0.555µs	attempt7.py
jiab_85281	27.061µs	+0.044µs	sub_sync_v3.py
nrehiew	27.155µs	+0.094µs	bestv2.py
sahanp	27.924µs	+0.770µs	gemv.py
qinking	28.088µs	+0.164µs	p1_v2.py
dxyz	28.219µs	+0.131µs	template_v1.py
guaguabear	28.292µs	+0.073µs	emplate_cute_bk_best.p
rkeldj	28.767µs	+0.475µs	code_0.py
CatsRCool	30.072µs	+1.305µs	s_attempt.py
kevin	30.261µs	+0.188µs	v2.py
houdinied	30.307µs	+0.046µs	submissionv3.py
sbstndbs	30.457µs	+0.150µs	submission_cute_3.py
mysfi	30.587µs	+0.130µs	submission_inline.py
John Hahn	30.596µs	+0.008µs	mid8.py
Clem_Hardy	30.622µs	+0.027µs	35.py
ozamatash	30.656µs	+0.033µs	submission.py
Rudrashis	31.931µs	+1.275µs	sub_rudrashis.py
gilsaia	32.270µs	+0.339µs	triton_4.py
Emilio	32.280µs	+0.010µs	kernel28.py
steve	32.343µs	+0.063µs	submission.py
currybab	32.475µs	+0.133µs	bmission_l1_specialized.
amackenzie-jumptrac	32.810µs	+0.335µs	submission.py
kurakura	33.257µs	+0.447µs	submission.py
lucifer_0000007	33.325µs	+0.068µs	submission.py
Alex A	34.763µs	+1.438µs	slow.py

tacowaco	34.792µs	+0.029µs	sub_cute_dsl_opt_7.py
moleabbu	35.328µs	+0.536µs	submission.py
revess	35.647µs	+0.319µs	submission2_staging.py
Arseni Ivanov	36.208µs	+0.561µs	ute_dsl_thread_blocked.p
Theta Sigma	36.856µs	+0.648µs	nvfp4_dom6.py
shiyegao	37.362µs	+0.506µs	template.py
shigao	37.408µs	+0.046µs	node_12.py
ethylene	37.716µs	+0.307µs	submission.py
__seal	37.759µs	+0.043µs	submission.py
bargav	37.773µs	+0.014µs	submission.py
michaelp5386	40.237µs	+2.464µs	gemv.py
Anurag	41.578µs	+1.341µs	submission.py
andrewpeterson	41.972µs	+0.394µs	submission3.py
cudawarped	42.514µs	+0.542µs	submission_cute2.py
d3bach	43.159µs	+0.645µs	nvfp4_gemv_cl_p_opt.py
tp6112	43.446µs	+0.287µs	submission.py
nullplayy	43.596µs	+0.149µs	cute_v2.py
akzaidan	43.895µs	+0.299µs	cute2.py
shiyeegao	43.912µs	+0.017µs	node_26.py
finalmouse	44.629µs	+0.717µs	cute_optim.py
sk4301	44.822µs	+0.193µs	e_submission_kernel_v6
taka09203	46.912µs	+2.090µs	gemv_cute__49MicroSec
Petr_Rocoss	48.507µs	+1.596µs	nission_hybrid_v4_reduc
.ryanrong	50.923µs	+2.416µs	39_pytorch_multi_batch
t_cc	53.025µs	+2.101µs	reference_cute_v2.py
Batuhanaktas	53.175µs	+0.150µs	i_am_not_a_dev_v4.py
drillyb	55.376µs	+2.202µs	submission.py
vadmear	60.034µs	+4.658µs	cute_gemv.py
wolfeheart	63.085µs	+3.051µs	submission_v148f.py
eashwar	64.374µs	+1.289µs	submission_3.py

Siazed	64.379µs	+0.005µs	submission.py
Jasmin Bogatinovski	64.487µs	+0.108µs	fourth_sub.py
tylerguest	64.490µs	+0.004µs	submission.py
georges314	64.986µs	+0.496µs	23_perm_scaledmm.py
Founzo	66.300µs	+1.314µs	submission_v3.py
shikhar	70.369µs	+4.068µs	fp4_gemv_v17_swizzled.
agokrani	70.469µs	+0.100µs	vfp4_gemv_v18_hybrid.p
Skoog	73.407µs	+2.938µs	cute_ref.py
maalvi	73.715µs	+0.309µs	s_submission.py
Light	75.202µs	+1.487µs	test_cute_v1.py
DizzleRama	76.658µs	+1.456µs	ıized_gemv_kernel_triton
ghadiaravi13	76.794µs	+0.136µs	template_cute.py
muxfd	79.909µs	+3.115µs	submission.py
John	80.078µs	+0.169µs	submission_v2.py
Rayleon	86.885µs	+6.807µs	cuda.py
건희	90.162µs	+3.277µs	baseline.py
phuc9702	92.829µs	+2.667µs	nvfp4_gemv.py
take000	97.952µs	+5.122µs	gemv.py
rbyrots	101.416µs	+3.464µs	submission_v1.py
RIM#0013	103.410µs	+1.994µs	new.py
inarikami	103.464µs	+0.055µs	template_tc_fix.py
roaringkitty67	103.525µs	+0.061µs	template_cute.py
pengwh	103.605µs	+0.080µs	template_cute.py
Zebz	103.679µs	+0.074µs	lesson1.py
mekkcyber	103.704µs	+0.025µs	template_cute_default.py
arb8020	103.737µs	+0.033µs	cute_kernel_template.py
someone_2	103.740µs	+0.004µs	sub-1.py
Siddhesh	103.745µs	+0.004µs	sec1.py
Smexy	103.769µs	+0.024µs	very_cute.py
jerry	103.769µs	+0.000µs	template_cute.py

DAWN	103.805µs	+0.036µs	submission.py
blink1205	103.860µs	+0.055µs	submission.py
MINJAE	103.861µs	+0.001µs	submission.py
konakona666	103.877µs	+0.016µs	template_cute.py
Naomath	103.895µs	+0.017µs	template_cute.py
Yeargod	103.899µs	+0.005µs	cute_baseline.py
houdiniedo	103.902µs	+0.003µs	template_cute.py
czhu_	103.917µs	+0.015µs	template_cute.py
dusu2	103.943µs	+0.026µs	submission.py
stargaryen622	103.977µs	+0.034µs	nvgemvfp4.py
hososuke	103.980µs	+0.004µs	template_cute.py
madhur	103.981µs	+0.001µs	submission.py
gearfive	104.008µs	+0.027µs	template_cute.py
whatdhack_	104.011µs	+0.003µs	submission.py
ayghri	104.019µs	+0.008µs	template_cute.py
manshar	104.070µs	+0.051µs	template_cute.py
phillass	104.166µs	+0.096µs	template_cute.py
chizuchizu	104.577µs	+0.411µs	template_cute.py
jihun_lee__	106.662µs	+2.085µs	template_cute.py
wkwkes9611	107.047µs	+0.386µs	template_cute.py
P	107.048µs	+0.001µs	template_cute.py
Harshil	108.542µs	+1.494µs	cutie.py
forestier_kasapi	116.542µs	+8.000µs	submission10.py
dpang	121.532µs	+4.990µs	sub3.py
arsrivish26691	126.408µs	+4.877µs	nvfp4tsk1vg.py
Nganga kamau	140.529µs	+14.121µs	submission.py
Jaber	142.263µs	+1.734µs	submission.py
fl4res	144.208µs	+1.945µs	fp4_nogcc.py
noobmaster69_og	150.843µs	+6.635µs	nission_pytorch_optimize
lucifer_is_back_	151.311µs	+0.468µs	submission.py

data_pi	151.527µs	+0.216µs	submission.py
esraa1602	151.578µs	+0.051µs	fsubmission.py
Xavier Init	151.785µs	+0.207µs	submission.py
abao_ai	151.864µs	+0.079µs	submission.py
Black BoB	151.878µs	+0.014µs	submission.py
ifndef_rust_define_ru	151.941µs	+0.064µs	nvfp4_gemv_cutlass4.py
Akhil	151.960µs	+0.019µs	submission.py
Tuan0Ge	152.047µs	+0.087µs	submission.py
stori	152.067µs	+0.020µs	test_submission.py
9thbeerr	152.085µs	+0.018µs	submission.py
dandelion4	152.431µs	+0.346µs	reference.py
blake	152.676µs	+0.245µs	sub1.py
poornaravuri	153.036µs	+0.361µs	submission.py
aazz061_33419	153.067µs	+0.031µs	submission.py
m.normansyah	153.146µs	+0.079µs	submission.py
jonters	153.599µs	+0.453µs	test_submission.py
Hamadullah Bijarani	153.606µs	+0.007µs	submission.py
arka26_24137	153.641µs	+0.034µs	submission.py
noahnisbet_70526	153.751µs	+0.110µs	submission.py
pcFaust	153.858µs	+0.107µs	better-call-ptx.py
flyingteemo2	154.241µs	+0.382µs	submission.py
akkolli	154.243µs	+0.002µs	submission.py
PunchShadow	154.294µs	+0.051µs	test_submission.py
mistobaan	154.340µs	+0.046µs	submission_vanilla.py
xXWarMachineRoXx	154.545µs	+0.206µs	submission.py
notjensenhuang	158.516µs	+3.971µs	p4_gemv_reference_CuTe
ago.lajko	171.314µs	+12.798µs	cuda2.py
vyom	186.345µs	+15.031µs	submission.py
elliotarledge	197.980µs	+11.635µs	v1.py
Fung	198.254µs	+0.274µs	submission_v2.py

xin	198.415µs	+0.161µs	sub.py
mobicham	198.509µs	+0.094µs	submission_dummy2.py
Sherlock	220.548µs	+22.039µs	submission.py
XoTic	225.321µs	+4.774µs	submission.py
kimhyeonho_xel	248.120µs	+22.799µs	submission.py
rt11	280.086µs	+31.965µs	v1.py
msuiche	312.000µs	+31.914µs	submission.py
sohail	313.816µs	+1.816µs	submission_tc_tmem.py
lyi	522.340µs	+208.524µs	submission.py
Infatoshi	746.905µs	+224.565µs	submission.py
albert9823	870.009µs	+123.104µs	submission_v2.py
leodaaaa	956.355µs	+86.346µs	submission-ld.py
anuragj0803	1036.500µs	+80.146µs	dia_problem1_submission
tmotmr	1457.411µs	+420.911µs	goodref.py
Tomas R.	1549.319µs	+91.908µs	submission.py
blurbird	1553.074µs	+3.755µs	submit.py
0x6c69676874	1558.692µs	+5.618µs	submission.py
tejesh06255	1571.334µs	+12.642µs	submission.py
runfme	1587.836µs	+16.502µs	submission.py
axeuron	1591.303µs	+3.467µs	submission.py
cherichy	1604.605µs	+13.302µs	reftorch.py
vinu1729	1704.280µs	+99.675µs	nvda.py
siro	1791.340µs	+87.060µs	submission.py
adityaljo123	1869.027µs	+77.688µs	submission.py
pank2025	1869.647µs	+0.620µs	mvfp4_gemv_baseline.py
freon	1870.278µs	+0.631µs	submission.py
lynn	1918.573µs	+48.295µs	submission.py
Joao	2133.069µs	+214.497µs	submission.py
Max Brashear	2236.387µs	+103.318µs	submission.py
rwxfortyseven	2352.616µs	+116.229µs	submission.py

XHN	2931.445µs	+578.829µs	submit_fp4.py
Voldemort4321	3027.244µs	+95.799µs	submission.py

Performance Trend

Contestant Search

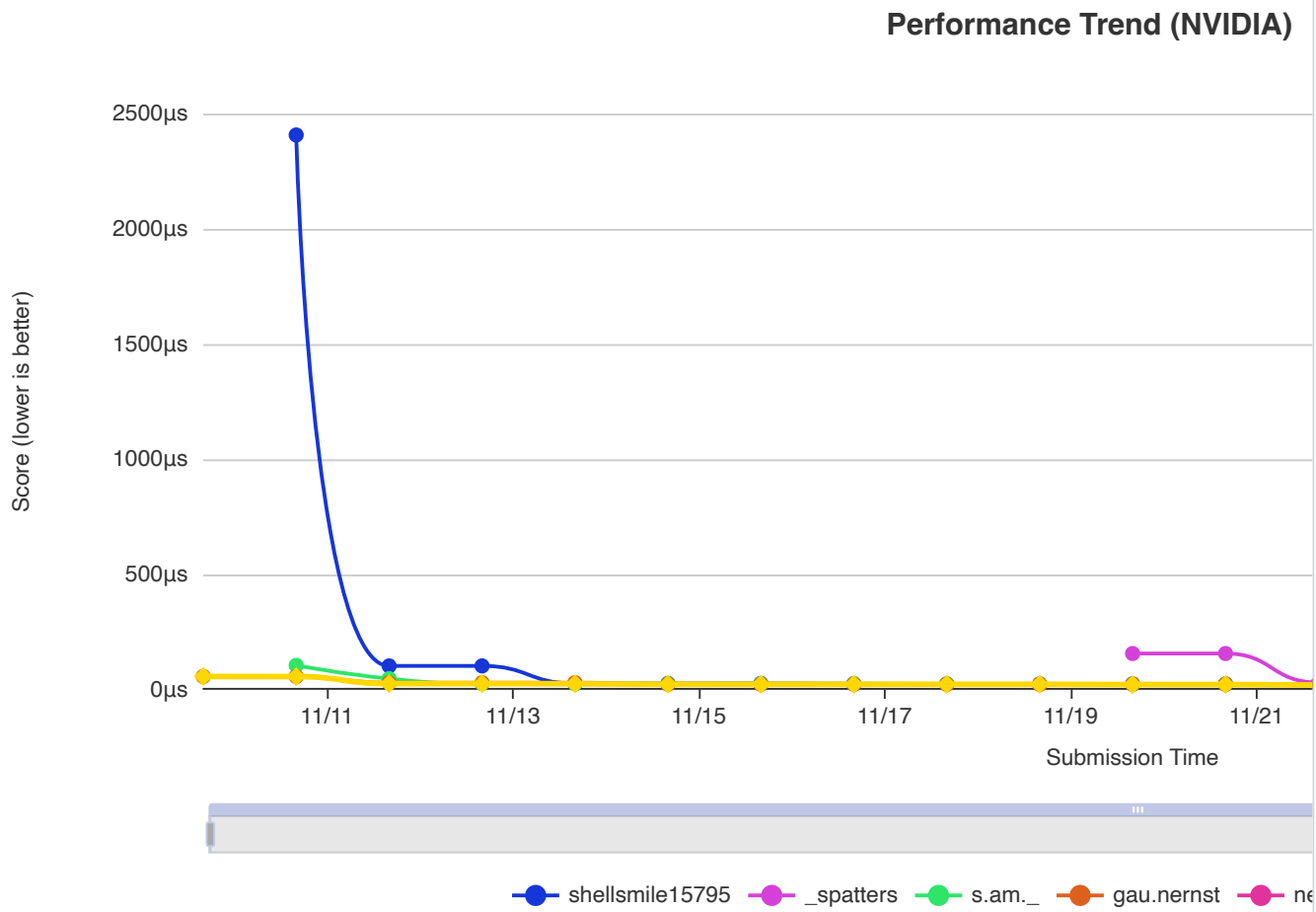
s.am._ gau.nernst shellsmile15795 _spatters netvope Type to search for con...

GPU Type: NVIDIA

Load Top 5

Best Over Time

Raw Submissions



X-Axis (Date): Start mm/dd/yyyy End mm/dd/yyyy Y-Axis (µs): Min Max Apply

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